Reproducing Table I of “Tracking down Quirks at the LHC”

Knapen, Lou, Papucci & Setford, arXiv:1708.02243 · paper-stated acceptance with an inferred longitudinal detector model · 3 October 2026

Setup

- **Stated in the paper or its detector reference [27]:** pixel and SCT barrel only ($|\eta| \lesssim 1.8$); pixel-centre hits; neighbouring pixels merged; SCT 34 μm across (2 × 17 μm); 45 mm beam spot; 25 ns timing; trigger pT(QQ) > 200 GeV; the paper's plane-finding algorithm and signal region.
- **Our reading (not stated):** SCT smearing along the beam of 1.16 mm (2 × the 0.58 mm in [27]); the 45 mm beam spot as one common shift per event, with the fit measured from the true vertex.
- **Also shown:** 0.58 mm (the value in [27], not doubled) and no smearing along the beam (no source).
- Two independent samples per benchmark (1000 and 2000 generated events), never combined. No tuning to the paper.

Definitions

ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(matched events)
ε_{fid} = N(triggered, every quirk crosses all 8 layers inside $|\eta| < 1.8$, within 25 ns) / N(triggered)
ε_{reco} = N(fiducial and plane found, 2 hits in 7 layers + ≥ 1 in the 8th, $\Delta w < 1$ cm, $\Delta s < 0.01$ cm) / N(fiducial)
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$ (paper Eq. 12)

Colored only: both R-hadrons must be charged; this cut is applied to the triggered events before ε_{fid} (27 of 101 and 68 of 207 events). The paper does not say which factor contains it.

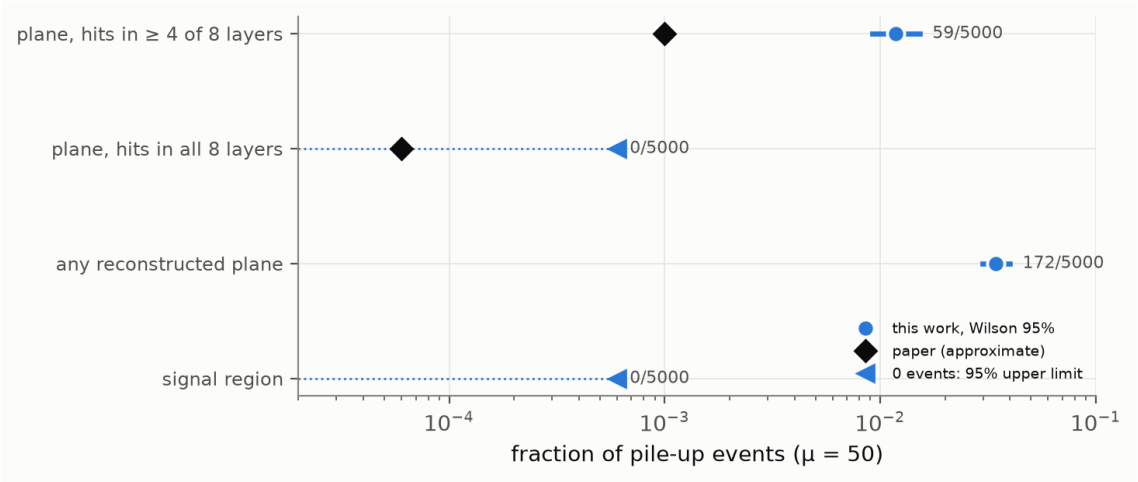
Standard error: $SE = \sqrt{\varepsilon (1 - \varepsilon) / n}$
Wilson 95% interval: $[\ \varepsilon + z^2/2n \pm z \sqrt{\varepsilon(1-\varepsilon)/n + z^2/4n^2}\ ] / (1 + z^2/n)$, $z = 1.96$
Upper limit (UL): when 0 of n events pass, the largest true rate that still gives 0 events in at least 5% of experiments: $UL = 1 - 0.05^{1/n} \approx 3/n$
p-value (exact two-sided binomial test): if the paper's value were exactly right, the probability of getting a count as unlikely as ours or more (the paper gives no counts or errors, so its own uncertainty is ignored)

p-value	Verdict
$p < 0.01$	**Strongly inconsistent**
$0.01 \le p < 0.05$	**Statistically inconsistent**
$0.05 \le p < 0.10$	**Borderline / mild tension**
$p \ge 0.10$	**No statistically significant inconsistency**

1. Pile-up background (⟨μ⟩ = 50, 5000 events)

Quantity	k/n	Rate ± SE	Wilson 95% / UL	Paper	Comparison
Plane, hits in ≥ 4 of 8 layers	59/5000	$1.2\times10^{-2} \pm 1.5\times10^{-3}$	$[9.2\times10^{-3}, 1.5\times10^{-2}]$	$\approx 1\times10^{-3}$	Higher than the paper's approximate rate
Plane, hits in all 8 layers	0/5000	0	UL 6.0×10^{-4}	$\approx 6\times10^{-5}$	Compatible with the paper's approximate rate
Any reconstructed plane	172/5000	$3.4\times10^{-2} \pm 2.6\times10^{-3}$	$[3.0\times10^{-2}, 4.0\times10^{-2}]$	—	—
Hit multiplicity	0/5000	0	UL 6.0×10^{-4}	—	No event passes
Signal region	0/5000	0	UL 6.0×10^{-4}	—	No event passes

The paper gives approximate rates without counts or errors, so no formal test is made.



Pile-up rates with Wilson 95% intervals; ◁ = 95% upper limit when no event passes; diamonds = paper.

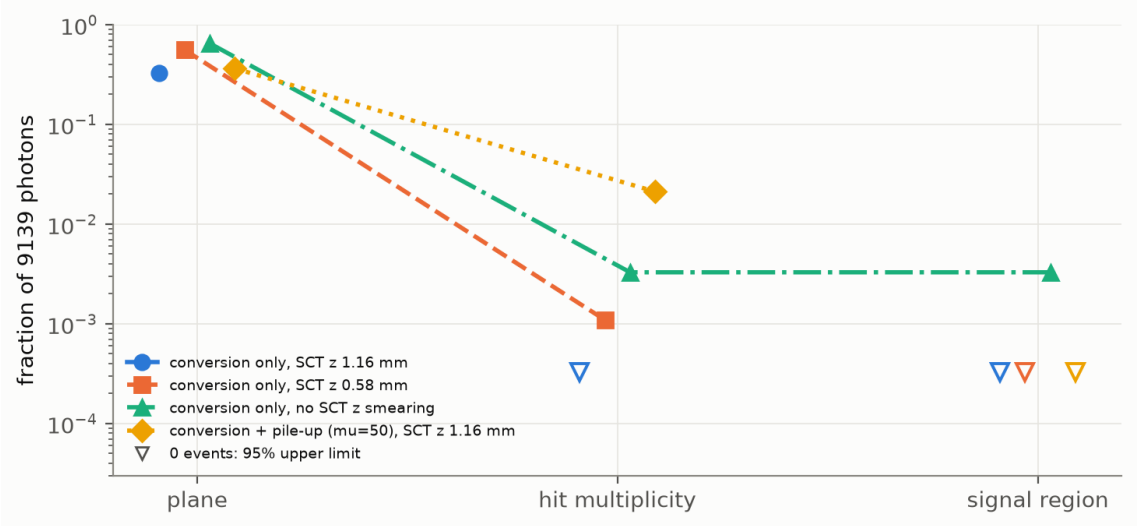
Reading

- The 4-of-8 rate is about 12× the paper's. Possible reasons, **not tested here**: our pile-up tracks are ideal helices with no energy loss, bremsstrahlung or endcaps; our pool has 20 000 minimum-bias events (paper 125 000); the paper does not define “a plane” for this number.
- “Compatible” for the all-8-layer rate and the signal region only means our 5000 events are not sensitive enough: the 95% upper limit (6×10^{-4}) is 10× above the paper's 6×10^{-5}.

2. Converted-photon background (9139 Z(νν)γ+jet photons, |η| < 1.8, fit from the vertex)

Setup	Plane	Hit mult.	Signal region	Wilson 95% / UL	Events at 300 fb⁻¹ (our σ / paper σ)
conversion only SCT z 1.16 mm	3005/9139	0/9139	0/9139	UL 3.3×10^{-4}	< 0.21 / < 0.98
conversion only SCT z 0.58 mm	5124/9139	10/9139	0/9139	UL 3.3×10^{-4}	< 0.21 / < 0.98
conversion only no SCT z smearing	5937/9139	30/9139	30/9139	$[2.3\times10^{-3}$, $4.7\times10^{-3}]$	2.1 / 9.8
conversion + pile-up (μ = 50) SCT z 1.16 mm	3323/9139	193/9139	0/9139	UL 3.3×10^{-4}	< 0.21 / < 0.98

Events at 300 fb⁻¹ = σ × 1% conversion probability × fraction × 300 fb⁻¹, with σ = 0.21 pb (our MadGraph) or ≈ 1 pb (paper). With 0 events the 95% upper limit is used.



Fraction of photons passing each cut; open triangles = 95% upper limit when no event passes.

Reading

- With SCT smearing along the beam (1.16 or 0.58 mm), no conversion reaches the signal region, with or without pile-up. The paper does not claim none passes: it notes that its Fig. 5 “suggests a few background events after the final selection cut”. In that figure 3.6% of the conversion Δs distribution is below 0.01 cm; with 1.16 mm, 3.7% of our conversion planes are, but none of them also passes the hit-count cut.
- Pile-up hits can fill the gaps of an $e^{+}e^{-}$ pair (193/9139 pass the hit count), but none of them is thin enough to pass Δs.
- Without SCT smearing along the beam, 30/9139 = 0.33% of conversions pass; then almost every conversion plane is thinner than the cut, unlike the paper's Fig. 5 (section 5).
- Our cross-section is 4.3× below the paper's ≈ 1 pb (generator settings not given). The 1% conversion probability is taken from the paper's own “≈ 1 pb → ≈ 10 fb”.

3. Colored quirks (m_Q = 1800 GeV, Λ = 2 keV)

Efficiency	Sample	k/n	ε ± SE	Wilson 95%	Paper	p	Verdict
ε_{trig}	COL1000	101/419	0.241 ± 0.021	[0.203, 0.284]	0.24	0.95	No statistically significant inconsistency
ε_{trig}	COL2000	207/873	0.237 ± 0.014	[0.210, 0.266]	0.24	0.87	No statistically significant inconsistency
ε_{fid}	COL1000	7/27	0.259 ± 0.084	[0.132, 0.447]	0.28	1.00	No statistically significant inconsistency
ε_{fid}	COL2000	25/68	0.368 ± 0.058	[0.263, 0.486]	0.28	0.14	No statistically significant inconsistency
ε_{reco}, SCT z 1.16 mm	COL1000	1/7	0.143 ± 0.132	[0.026, 0.513]	0.35	0.43	No statistically significant inconsistency
ε_{reco}, SCT z 1.16 mm	COL2000	3/25	0.120 ± 0.065	[0.042, 0.300]	0.35	0.019	Statistically inconsistent
ε_{reco}, SCT z 0.58 mm	COL1000	3/7	0.429 ± 0.187	[0.158, 0.750]	0.35	0.70	No statistically significant inconsistency
ε_{reco}, SCT z 0.58 mm	COL2000	5/25	0.200 ± 0.080	[0.089, 0.391]	0.35	0.14	No statistically significant inconsistency
ε_{reco}, SCT z none	COL1000	5/7	0.714 ± 0.171	[0.359, 0.918]	0.35	0.056	Borderline / mild tension
ε_{reco}, SCT z none	COL2000	11/25	0.440 ± 0.099	[0.267, 0.629]	0.35	0.40	No statistically significant inconsistency

4. Electroweak quirks (m_Q = 800 GeV, Λ = 4 keV)

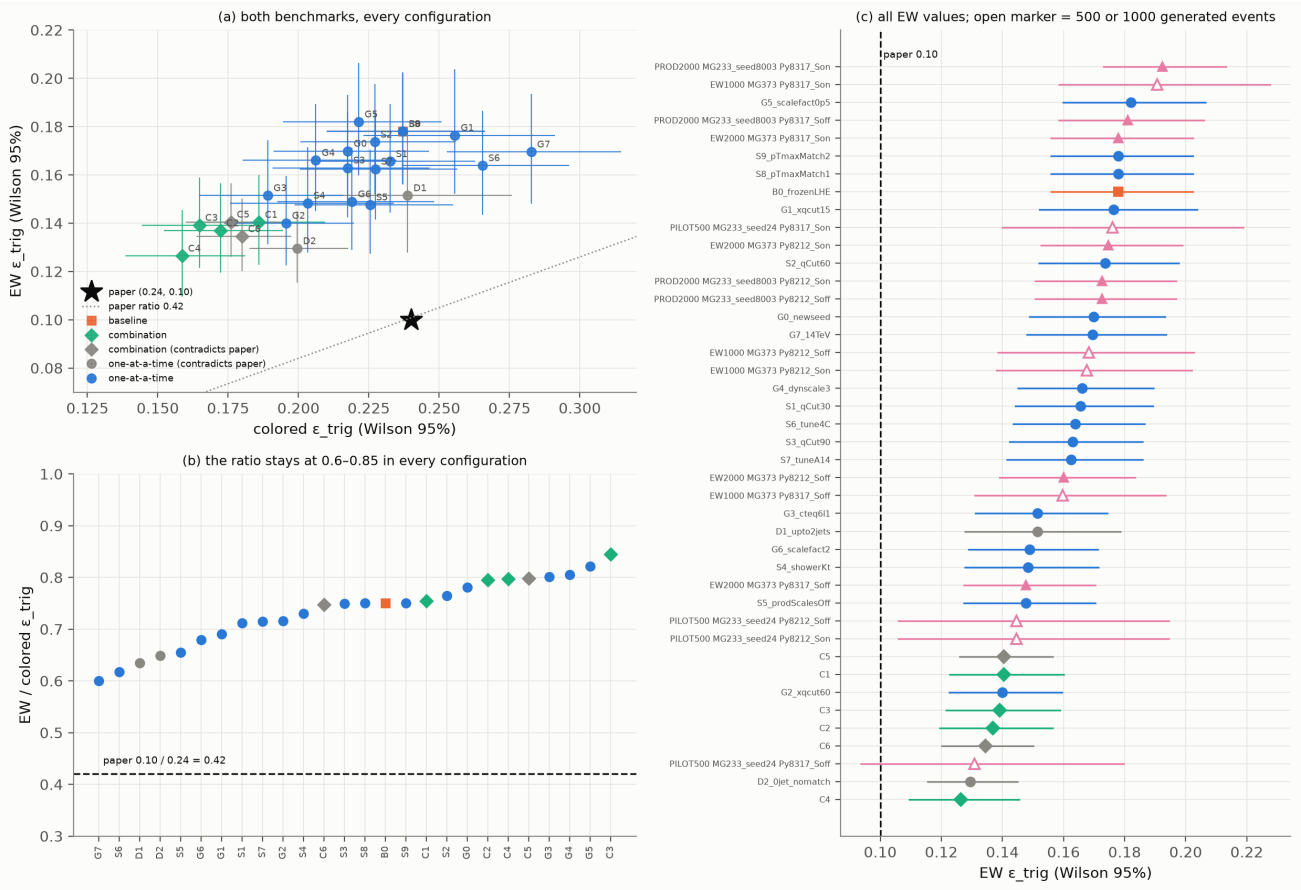
Efficiency	Sample	k/n	ε ± SE	Wilson 95%	Paper	p	Verdict
ε_{trig}	EW1000	89/511	0.174 ± 0.017	[0.144, 0.209]	0.10	2.8×10^{-7}	Strongly inconsistent
ε_{trig}	EW2000	186/1045	0.178 ± 0.012	[0.156, 0.202]	0.10	1.8×10^{-14}	Strongly inconsistent
ε_{fid}	EW1000	40/89	0.449 ± 0.053	[0.350, 0.553]	0.28	8.2×10^{-4}	Strongly inconsistent
ε_{fid}	EW2000	94/186	0.505 ± 0.037	[0.434, 0.576]	0.28	1.3×10^{-10}	Strongly inconsistent
ε_{reco}, SCT z 1.16 mm	EW1000	6/40	0.150 ± 0.056	[0.071, 0.291]	0.72	9.1×10^{-14}	Strongly inconsistent
ε_{reco}, SCT z 1.16 mm	EW2000	21/94	0.223 ± 0.043	[0.151, 0.318]	0.72	2.4×10^{-23}	Strongly inconsistent
ε_{reco}, SCT z 0.58 mm	EW1000	10/40	0.250 ± 0.068	[0.142, 0.402]	0.72	9.4×10^{-10}	Strongly inconsistent
ε_{reco}, SCT z 0.58 mm	EW2000	29/94	0.309 ± 0.048	[0.224, 0.408]	0.72	1.5×10^{-16}	Strongly inconsistent
ε_{reco}, SCT z none	EW1000	34/40	0.850 ± 0.056	[0.709, 0.929]	0.72	0.078	Borderline / mild tension
ε_{reco}, SCT z none	EW2000	76/94	0.809 ± 0.041	[0.717, 0.875]	0.72	0.065	Borderline / mild tension

Reading

- **Colored:** trigger and fiducial agree with the paper. Reconstruction agrees for 0.58 mm and for no smearing; at 1.16 mm the larger sample is low (3/25).
- **EW trigger:** 17–18% against 10%. The same definition gives the paper's colored value. 36 one-at-a-time generator variations and 6 pre-registered combinations of the settings that lowered it most keep EW at 12.6–19%; every combination also lowers colored to 16–19%, so the EW/colored ratio stays at 0.60–0.85 against the paper's 0.42 (table and figure below). The paper's generator settings are not given.
- **EW fiducial:** 45–51% against 28%. Using the detector lengths of [27] instead of $|\eta| < 1.8$ gives 33–37%, close to 28% (section 6), but the paper does not say it used them.
- **EW reconstruction:** 15–31% with SCT smearing along the beam; 81–85% only without it.

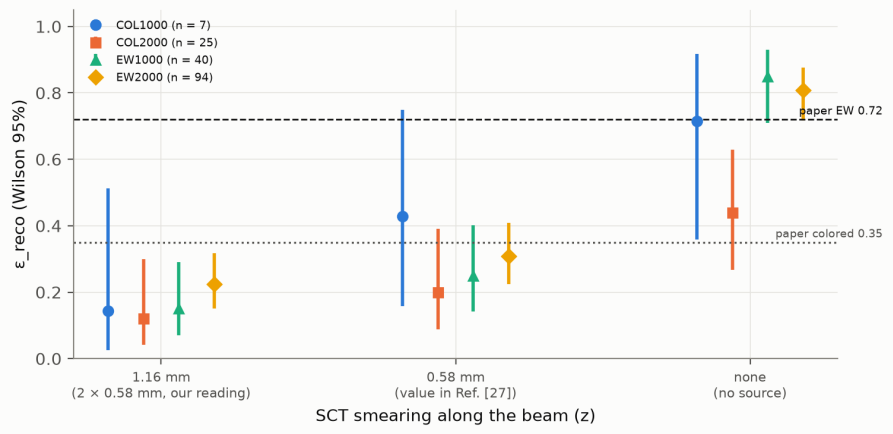
Trigger: pre-registered combinations (written down and hashed before running; 2000 generated events per benchmark; pass = paper inside Wilson 95% for both)

Combination	EW: k/n, ε [Wilson 95%]	Colored: k/n, ε [Wilson 95%]	EW/col.	Both pass?
C1: xqcut 60 + cteq6l1 + scale ×2	188/1339 = 0.140 [0.123, 0.160]	215/1156 = 0.186 [0.165, 0.209]	0.75	no
C2: C1 + production scales off	181/1322 = 0.137 [0.119, 0.156]	209/1213 = 0.172 [0.152, 0.195]	0.79	no
C3: C1 + showerKt	183/1315 = 0.139 [0.122, 0.159]	188/1141 = 0.165 [0.144, 0.187]	0.84	no
C4: C1 + showerKt + production scales off	167/1321 = 0.126 [0.110, 0.145]	179/1128 = 0.159 [0.139, 0.181]	0.80	no
C5: 0 jets, no matching + cteq6l1 + scale ×2 (contradicts paper)	281/2000 = 0.141 [0.126, 0.156]	352/2000 = 0.176 [0.160, 0.193]	0.80	no
C6: C5 + production scales off (contradicts paper)	269/2000 = 0.135 [0.120, 0.150]	360/2000 = 0.180 [0.164, 0.197]	0.75	no

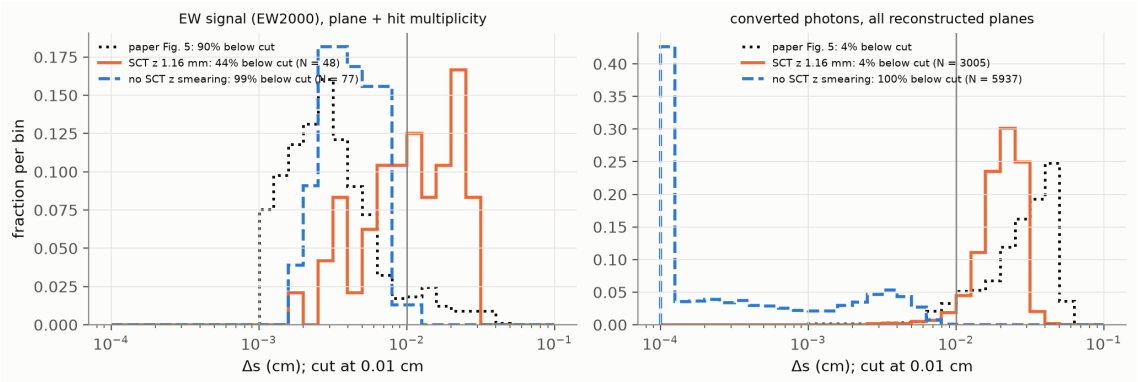


(a) EW vs colored ε_trig for every configuration run on both benchmarks, labelled (blue = one-at-a-time, green = combinations, grey = settings that contradict the paper's 'up to 1 jet, matched', orange = baseline, star = paper). (b) EW/colored ratio per configuration. (c) All EW values incl. the generator-version pilot (pink); open markers = 500 or 1000 events.

5. What decides reconstruction



Reconstruction efficiency of the same events for each SCT smearing along the beam. Bars = Wilson 95%; lines = paper.



Δs against the paper's Fig. 5: EW signal (left) and converted photons (right), with 1.16 mm SCT smearing along the beam (solid) and without it (dashed).

Can the paper's own figures pick the SCT setting?

Paper observable	Paper	SCT z 1.16 mm	0.58 mm	none
EW reconstruction (EW2000)	0.72	0.22	0.31	0.81
Colored reconstruction (COL2000)	0.35	0.12	0.20	0.44
EW signal Δs below the cut	90%	44%	39%	99%
EW signal Δs median	≈ 28 µm	110 µm	121 µm	41 µm
Conversion Δs below the cut (all planes)	3.6%	3.7%	5.5%	100.0%
Conversion Δs median (all planes)	≈ 282 µm	211 µm	185 µm	2 µm

Reading

- [27] gives the SCT accuracy as 17 µm across and 0.58 mm along the beam. The paper doubles “the intrinsic accuracy” without naming the direction, so 1.16 mm is our reading of the text.
- The signal rows favour no smearing along the beam: the signal planes get thin enough and the efficiencies come close to Table I.
- The conversion rows favour 1.16 mm: the paper's conversion Δs (3.6% below the cut, median ≈ 280 µm) matches 1.16 mm (3.7%, 211 µm), while without smearing almost every conversion plane is far thinner than the cut.
- So no single SCT setting reproduces both panels of the paper's Fig. 5 in our simulation. The paper's figures cannot decide the setting; we keep all three treatments.

45 mm beam spot. The paper smears “the z-coordinates of all the hits” and assumes the vertex “is identified correctly, and is located at the origin”. Events passing the signal region for SCT z 1.16 · 0.58 mm · none:

Reading of the beam-spot sentence	Colored 2000	EW 2000
One common shift per event, fit from the true vertex (used)	3 · 5 · 11 of 25	21 · 29 · 76 of 94
One common shift per event, fit from the origin	0 · 0 · 0 of 25	3 · 3 · 4 of 94
Separate 45 mm shift of every hit	0 · 0 · 0 of 21	0 · 0 · 0 of 66

6. Table I: paper vs this work

	Quantity	Paper	1000-event sample: ε ± SE, Wilson 95%, p	2000-event sample: ε ± SE, Wilson 95%, p	Verdict (2000)
Colored	ε_{trig}	0.24	101/419 = 0.241 ± 0.021 [0.203, 0.284], p = 0.95	207/873 = 0.237 ± 0.014 [0.210, 0.266], p = 0.87	**No statistically significant inconsistency**
	ε_{fid}	0.28	7/27 = 0.259 ± 0.084 [0.132, 0.447], p = 1.00	25/68 = 0.368 ± 0.058 [0.263, 0.486], p = 0.14	**No statistically significant inconsistency**
	ε_{reco}, z 1.16 mm	0.35	1/7 = 0.143 ± 0.132 [0.026, 0.513], p = 0.43	3/25 = 0.120 ± 0.065 [0.042, 0.300], p = 0.019	**Statistically inconsistent**
	ε_{reco}, z 0.58 mm	0.35	3/7 = 0.429 ± 0.187 [0.158, 0.750], p = 0.70	5/25 = 0.200 ± 0.080 [0.089, 0.391], p = 0.14	**No statistically significant inconsistency**
	ε_{reco}, no z smear	0.35	5/7 = 0.714 ± 0.171 [0.359, 0.918], p = 0.056	11/25 = 0.440 ± 0.099 [0.267, 0.629], p = 0.40	**No statistically significant inconsistency**
EW	ε_{trig}	0.10	89/511 = 0.174 ± 0.017 [0.144, 0.209], p = 2.8×10^{-7}	186/1045 = 0.178 ± 0.012 [0.156, 0.202], p = 1.8×10^{-14}	**Strongly inconsistent**
	ε_{fid}	0.28	40/89 = 0.449 ± 0.053 [0.350, 0.553], p = 8.2×10^{-4}	94/186 = 0.505 ± 0.037 [0.434, 0.576], p = 1.3×10^{-10}	**Strongly inconsistent**
	ε_{reco}, z 1.16 mm	0.72	6/40 = 0.150 ± 0.056 [0.071, 0.291], p = 9.1×10^{-14}	21/94 = 0.223 ± 0.043 [0.151, 0.318], p = 2.4×10^{-23}	**Strongly inconsistent**
	ε_{reco}, z 0.58 mm	0.72	10/40 = 0.250 ± 0.068 [0.142, 0.402], p = 9.4×10^{-10}	29/94 = 0.309 ± 0.048 [0.224, 0.408], p = 1.5×10^{-16}	**Strongly inconsistent**
	ε_{reco}, no z smear	0.72	34/40 = 0.850 ± 0.056 [0.709, 0.929], p = 0.078	76/94 = 0.809 ± 0.041 [0.717, 0.875], p = 0.065	**Borderline / mild tension**

Bottom line

- **Agrees with the paper:** colored trigger and fiducial efficiency; colored reconstruction for 0.58 mm or no SCT smearing along the beam.
- **Does not agree:** EW trigger (0.18 vs 0.10, also with combined generator settings), EW fiducial (0.51 vs 0.28), and EW reconstruction with SCT smearing along the beam (0.22–0.31 vs 0.72).
- **Backgrounds:** pile-up only and conversion only give no signal-region event, and so does the conversion + pile-up overlay with this setup. Without SCT smearing along the beam, 0.33% of conversions would pass, and the conversion Δs shape would no longer match the paper's Fig. 5.
- **Why a full reproduction is not possible:** the paper does not fix the SCT smearing along the beam, the beam-spot treatment, the exact acceptance or the generator settings; its own Fig. 5 is not consistent with any single SCT setting in our simulation; and it gives no event counts or errors.

Sensitivity check: detector lengths from [27] (pixel $|z| < 40.05$ cm, SCT $|z| < 74.9$ cm) instead of $|\eta| < 1.8$

Sample	ε_fid = k/n ± SE	Wilson 95%	p vs 0.28	Signal region (z 1.16 · 0.58 mm · none)
COL1000	6/27 = 0.222 ± 0.080	[0.106, 0.408]	0.67	1 · 2 · 5 of 6
COL2000	22/68 = 0.324 ± 0.057	[0.224, 0.442]	0.42	1 · 3 · 8 of 22
EW1000	33/89 = 0.371 ± 0.051	[0.278, 0.475]	0.059	2 · 5 · 27 of 33
EW2000	62/186 = 0.333 ± 0.035	[0.270, 0.404]	0.12	10 · 14 · 50 of 62

Appendix A. Where every setting comes from

Green = stated in the paper or [27]; amber = our reading; red = no source. Counts are listed in the order COL1000, COL2000, EW1000, EW2000.

Setting	Source	Status	Resulting counts
Barrel only (pixel + SCT, 8 layers)	Paper Sec. III A: 'we only included the barrel of the pixel and SCT detectors in our simulations'	**PAPER-STATED**	frame for all counts
Acceptance: effective |eta| < 1.8, per hit, no finite-z boundary	Paper Sec. III A: '... which effectively restricts the fiducial range to |eta| <~ 1.8' (applying it as a per-hit cut is the implementation)	**PAPER-STATED**	fiducial COL1000 7/27, COL2000 25/68, EW1000 40/89, EW2000 94/186
Acceptance: finite barrel lengths (pixel |z|<400.5 mm, SCT |z|<749 mm, IBL |z|<330.15 mm)	Ref. [27] (arXiv:0901.0512 pp. 40-43) table 'Sensitive barrel 0 < |z| < 400.5 / 749'; IBL arXiv:1803.00844. No evidence the authors applied them	**INFERRED (sensitivity)**	fiducial COL1000 6/27, COL2000 22/68, EW1000 33/89, EW2000 62/186
Layer radii IBL 33.25, pixel 50.5/88.5/122.5, SCT 299/371/443/514 mm	Ref. [27] ranges '50.5<R<122.5', '299<R<514'; ATLAS JINST 3 S08003; IBL from arXiv:1803.00844 and paper Fig. 4 spacing	**REFERENCE-STATED (IBL INFERRED)**	all counts
Pixel hit = centre of hit pixel, 50x400 um (IBL 50x250 um)	Paper: 'the center of each pixel hit is used', 'pixel size rather than the resolution'; Ref. [27]: 'minimum pixel size on a sensor is 50x400 um2'	**PAPER-STATED + REFERENCE-STATED (IBL pitch INFERRED)**	all counts
Neighbouring-pixel merging	Paper: 'Hits in neightboring pixels ... are merged to a single hit' (adjacency rule INFERRED)	**PAPER-STATED**	all counts
SCT R-phi resolution 34 um	Paper: 'we conservatively assume a resolution of twice the intrinsic accuracy quoted in [27]'; Ref. [27]: '17 um (R-phi)'	**PAPER-STATED rule x REFERENCE-STATED value**	all counts
SCT z resolution 1160 um (= 2 x 580 um)	Same paper sentence (no coordinate named); Ref. [27]: 'intrinsic accuracies per module in the barrel are 17 um (R-phi) and 580 um (z)', footnote 'Arises from the 40 mrad stereo angle'	**INFERRED (paper rule applied to the reference z value; contradicted by paper Fig. 5)**	final COL1000 1/7, COL2000 3/25, EW1000 6/40, EW2000 21/94
SCT z resolution 580 um (1 x Ref. [27])	Ref. [27] value without the paper's doubling	**INFERRED (sensitivity; contradicts 'twice')**	final COL1000 3/7, COL2000 5/25, EW1000 10/40, EW2000 29/94
No SCT z measurement smearing	No source; closer to paper Fig. 5 Delta_s	**UNDOCUMENTED**	final COL1000 5/7, COL2000 11/25, EW1000 34/40, EW2000 76/94
45 mm longitudinal beam-spot smearing exists	Paper: 'a uniform, gaussian smearing with width 45 mm to the z-coordinates of all the hits'	**PAPER-STATED**	—
Beam spot = one common z shift per event, fit relative to the true vertex	Same sentence + 'We will assume that the primary vertex is identified correctly'	**INFERRED (wording ambiguous)**	SCT z 1160: COL2000 3/25, EW2000 21/94
Beam spot = one common z shift per event, fit relative to the origin	'... and is located at the origin' read literally	**INFERRED (alternative reading)**	SCT z 1160: COL2000 0/25, EW2000 3/94
Beam spot = independent 45 mm shift of every hit	'smearing ... to the z-coordinates of all the hits' read per hit	**INFERRED (alternative reading)**	fiducial COL2000 21/68, EW2000 66/186; final 0 and 0 for every SCT z (no plane found)
25 ns timing cut	Paper Sec. IV (location of the cut INFERRED)	**PAPER-STATED**	all counts
Delta_w = sqrt(lambda_2) (Eq. 11)	Paper Eq. 11; paper Fig. 4 band corresponds to 0.5 sqrt(lambda_2)	**PAPER-STATED**	no count changes
Trigger: pT(QQbar) > 200 GeV over matched events	Paper Sec. III B	**PAPER-STATED**	COL1000 101/419, COL2000 207/873, EW1000 89/511, EW2000 186/1045

Setting	Source	Status	Resulting counts
Generator settings (matching scale, PDF, scales, tune)	not stated in the paper	**UNDOCUMENTED**	EW trigger 0.13-0.19 over 36 single variations; 0.126-0.141 over 6 combinations (colored then 0.16-0.19)
Pile-up: mu = 50, 45 mm per vertex, ATLAS tracking-efficiency removal, pixel merging	Paper Sec. III D; ATL-PHYS-PUB-2015-051; pool 20 000 events (paper: 125 000)	**PAPER-STATED (pool size differs)**	signal region 0/5000
Converted photon: conversion in the beam pipe, equal energy sharing, 1% probability; overlay with pile-up	Paper Sec. III D (1% derived from '~1 pb -> ~10 fb')	**PAPER-STATED**	signal region 0/9139 (z 1.16), 0/9139 (0.58), 30/9139 (none); with pile-up 0/9139

Appendix B. Paper-figure reproductions

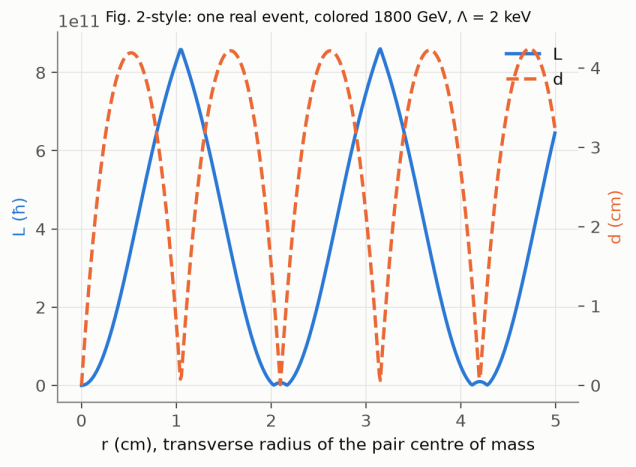


Fig. 2 style: angular momentum L and quirk separation d for one colored event (COL2000 event 107). L peaks near 9×10^{11} ħ, the paper's $\approx 10^{12}$ ħ scale.

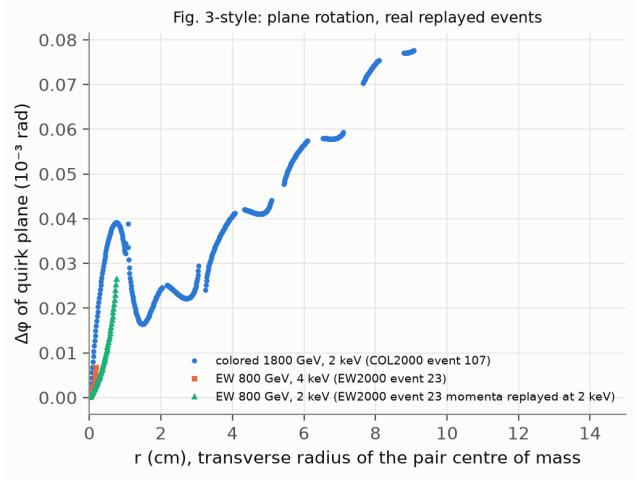


Fig. 3 style: the quirk plane rotates by less than 10^{-4} rad, so the hits stay on one plane.

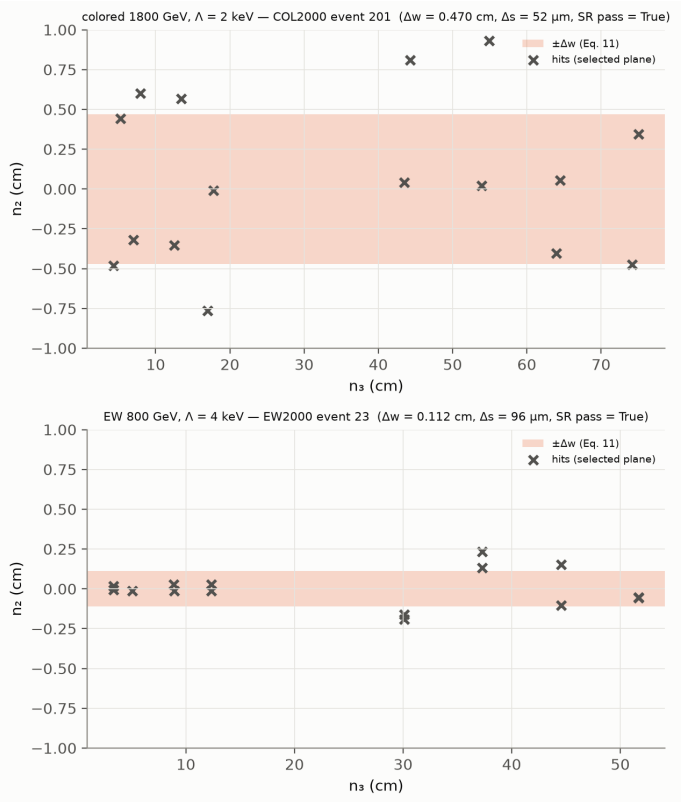


Fig. 4 style: hits of one colored (COL2000 event 201) and one EW (EW2000 event 23) event in the plane's own coordinates, SCT z 1.16 mm. Both pass the signal region (Δs = 52 and 96 μm). Band = ±Δw.